

VBF-C1T1R1 bom - Virtual Battery Factory deliverable

BOM - VBF-C1T1R1-BOM-001

[Sheet: BOM]

Item | Component | Parameter | Champion value | Unit | Source

1 | Current collector (+) | Positive current collector thickness [m] | 8e-06 | m | override work/par
2 | Current collector (-) | Negative current collector thickness [m] | 6e-06 | m | override (base 12
3 | Separator | Separator thickness [m] | 8e-06 | m | override (base 12e-6)
4 | Separator | Separator porosity | 0.55 | - | override (base 0.47)
5 | Electrolyte | Cation transference number | 0.7 | - | override (base 0.2594)
6 | Electrolyte | Electrolyte diffusivity [m².s⁻¹] | 2e-09 | m²/s | override (base Nyman2008 fn)
7 | Electrolyte | Electrolyte conductivity [S.m⁻¹] | 3 | S/m | override (base Nyman2008 fn)
8 | Negative electrode | Negative particle radius [m] | 3e-06 | m | override (base 5.86e-6)
9 | Thermal management | Total heat transfer coefficient [W.m⁻².K⁻¹] | 100 | W/m²/K | override (base
10 | Positive electrode | Positive electrode thickness [m] | 7.56e-05 | m | Chen2020 base (unchanged
11 | Negative electrode | Negative electrode thickness [m] | 8.52e-05 | m | Chen2020 base (unchanged
12 | Positive electrode | Positive particle radius [m] | 5.22e-06 | m | Chen2020 base (unchanged)
13 | Cell geometry | Electrode area [m²] | 0.1027 | m² | calc-energy output (0.065 x 1.58 m)